SURVEY RESEARCH PORTFOLIO · FAYE NIU

Designing and validating a 40-item questionnaire on graduate students' psychosocial development

Role Sole researcher, from design to write-up **Context** BSc thesis, Xi'an Technological University, 2024–25; re-analysed in 2026

Tools Wenjuanxing · SPSS · AMOS · Python · Excel · MATLAB

Research question. How are master's students doing across four areas of psychosocial development (mental health, social adaptation, academic pressure and self-actualisation), and can a short self-report questionnaire reliably sort them into risk levels that counsellors can act on?

Scope. Four constructs were defined from the literature, and items adapted from established scales formed a 40-item Likert questionnaire. It was fielded online to master's students (175 responses, 154 after cleaning) and tested for reliability and validity, and a weighted scoring model classified each respondent. For this portfolio I re-ran every analysis in Python to check whether the original conclusions hold.

154	.91–.92	.052	87–97%
usable responses out of 175 (88%)	Cronbach's α for each of the four subscales	RMSEA of the confirmatory factor model	classifications unchanged under alternative weights

Main result of the re-analysis

The thesis concluded that the four-dimension structure was valid because the standard fit indices passed. The re-analysis shows the four dimensions are statistically indistinguishable (latent correlations ≈ 1.0): the questionnaire reliably measures one overall level of psychosocial difficulty, not four separate ones. I therefore report a single total score and treat the subscales as descriptive. The same result explains why the scoring model barely depends on its weights (§5).

Workflow and tools

Step	Work	Tool	See
Design	Literature review, construct definition, item adaptation, questionnaire build	Wenjuanxing	§1–2
Data preparation	Cleaning, coding, block totals, descriptives and sample profile	SPSS, Excel	§3
Exploratory analysis	KMO and Bartlett's test, exploratory factor analysis and item removal, Cronbach's α	SPSS	§4
Confirmatory analysis	Four-factor CFA; χ^2/df, RMSEA, RMR, GFI, TLI, CFI	AMOS	§4
Scoring model	Weights, trapezoidal membership functions, two-level fuzzy evaluation	Python, Excel	§5
Prediction model	Normalisation, BP neural network, genetic-algorithm optimisation	MATLAB	App. B
Re-analysis (2026)	Data-quality screen, item analysis, parallel analysis, model comparison, weight sensitivity, segment tests, short form	Python	§3–8

The thesis also included a MATLAB neural-network model (GA-BP), summarised with a 2026 review in Appendix B. The re-analysis code is on GitHub: github.com/Newnew-niu/survey-instrument-validation.

1 · From research question to constructs

Most existing research on graduate students covers depression or anxiety screening; much less looks at psychosocial development more broadly. I used Maslow's hierarchy of needs as the framework, merging its five levels into four constructs suited to graduate-student life, each based on an established instrument:

Construct	Maslow level	Instrument adapted	Items
Mental health	Physiological + safety	Zung Self-Rating Depression Scale (SDS)	10
Social adaptation	Belonging	Need to Belong scale (DeWall et al.)	10
Academic pressure	Esteem	Academic Stress Questionnaire (Nwadiani); Postgraduate Learning Stress Questionnaire (Liu)	10
Self-actualisation	Self-actualisation	Short Index of Self-Actualization (Crandall et al.)	10

The design implies three testable claims:

1. The 10 items in each block measure one coherent construct (reliability).
2. The four constructs are distinct from each other (discriminant validity).
3. The combined score separates students into meaningful levels (scoring model).

2 · Questionnaire design

Introduction
Purpose, anonymity, reference period: "your time as a master's student"

→

Blocks A–D
4 × 10 statements, one construct per block

→

Profile
Gender and year of study, placed last to avoid priming

- **Scale:** a fully labelled 5-point agreement scale (1 = strongly disagree … 5 = strongly agree), the same for all 40 items.
- **Scoring:** every item is scored so that higher means more difficulty, so each block can simply be summed.
- **Adaptation:** items from clinical and workplace scales were reworded for students, for example "work" → "study" and "colleagues" → "classmates".

Wording review (2026)

A review of the item wording found several items I would now write differently:

Issue	Example	Change
Double-barrelled	A9 "…I feel exhausted and short of breath"	Split into two items, or keep only the core symptom
Off-construct	C4 "I find study and research dull", in the pressure block	Reword around pressure, or move it to an engagement measure
One-directional	All 40 items scored towards difficulty	Add 3–4 reverse-scored items per block
No quality checks	No attention or consistency items	Add one instructed-response item and one repeated item

3 · Fieldwork and data quality

The survey was built on Wenjuanxing and distributed to currently enrolled master's students. Of 175 responses,

Sample	n	%

154 (88%) were retained after removing incomplete and obviously careless ones.

The exclusion rules were not documented at the time. For the re-analysis, a documented screen was applied to the 154 retained cases:

Male	71	46.1
Female	83	53.9
Year 1	44	28.6
Year 2	56	36.4
Year 3	54	35.1

Check	Rule	Result
Missing data	Any blank item	**0 cases**
Straight-lining	Same answer on all 40 items (SD = 0)	**0 cases**
Long-string	20 or more identical answers in a row	**0 cases (longest run: 12)**
Multivariate outliers	Mahalanobis distance, $p < .001$	4 cases, kept; results unchanged when excluded

4 · Reliability and validity

Step 1 · Items and reliability

Subscale	Cronbach's α	KMO	Lowest item–total r	Composite reliability	AVE
Mental health	.911	.932	.60	.912	.509
Social adaptation	.918	.942	.65	.920	.534
Academic pressure	.918	.931	.59	.918	.530
Self-actualisation	.911	.936	.63	.911	.505

Thresholds used: α and composite reliability ≥ .70; KMO ≥ .70; item–total r ≥ .40; AVE ≥ .50. Deleting any single item did not raise α, and every item separated the top and bottom 27% of scorers (t ≥ 9.07).

Step 2 · Model fit

A four-factor confirmatory factor analysis (CFA; AMOS, replicated in Python) met conventional cut-offs: $\chi^2/df = 1.41$, RMSEA = .052, CFI = .93, TLI = .93, with standardised loadings between .63 and .81. The thesis treated this as sufficient evidence of validity; Step 3 shows it was not.

Step 3 · Discriminant validity

In the same CFA, the correlations between the four latent factors were 0.99–1.02, well above the ~.85 limit for discriminant validity. Three further checks pointed the same way:

- Parallel analysis supported 1 factor. The first eigenvalue alone explained 53% of the variance.
- A one-factor model fitted as well as the four-factor model and had the lower AIC and BIC (table below).
- In a bifactor model, the items loaded on the general factor (mean loading .72). The four specific factors added almost nothing (mean loading .05).

Model	χ^2/df	CFI	TLI	RMSEA	AIC	BIC
One general factor	1.43	.926	.922	.053	**146**	**389**
Four correlated factors (thesis)	1.41	.929	.925	.052	159	420
Bifactor (general + 4 specific)	1.44	.928	.920	.053	227	591

Conclusion

The questionnaire is a reliable measure of overall psychosocial difficulty. The 40-item total should be the main score, with the four subscale scores reported only as descriptives and not compared with each other. To measure the four areas separately, the item pools would need to be rewritten (§8).

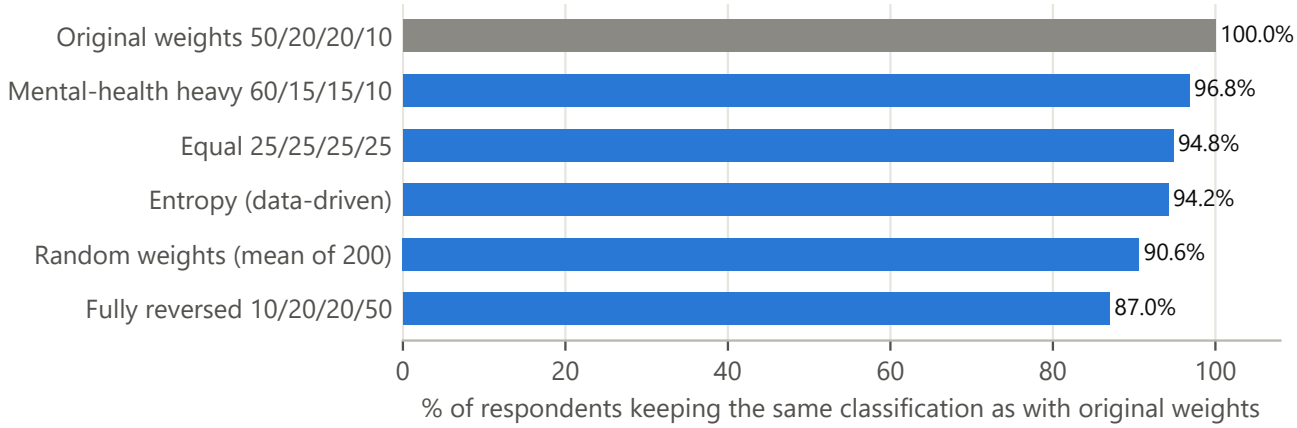
5 · Scoring model

To give counsellors categories rather than raw scores, I built a two-level fuzzy comprehensive evaluation model:

1. Each subscale total (range 10–50) is mapped by trapezoidal membership functions onto four levels: Good, Normal, Mild concern and Severe concern.
2. The four subscales are combined with weights based on Maslow's hierarchy: mental health 50%, social adaptation 20%, academic pressure 20%, self-actualisation 10%.
3. Each respondent is assigned the level with the highest membership.

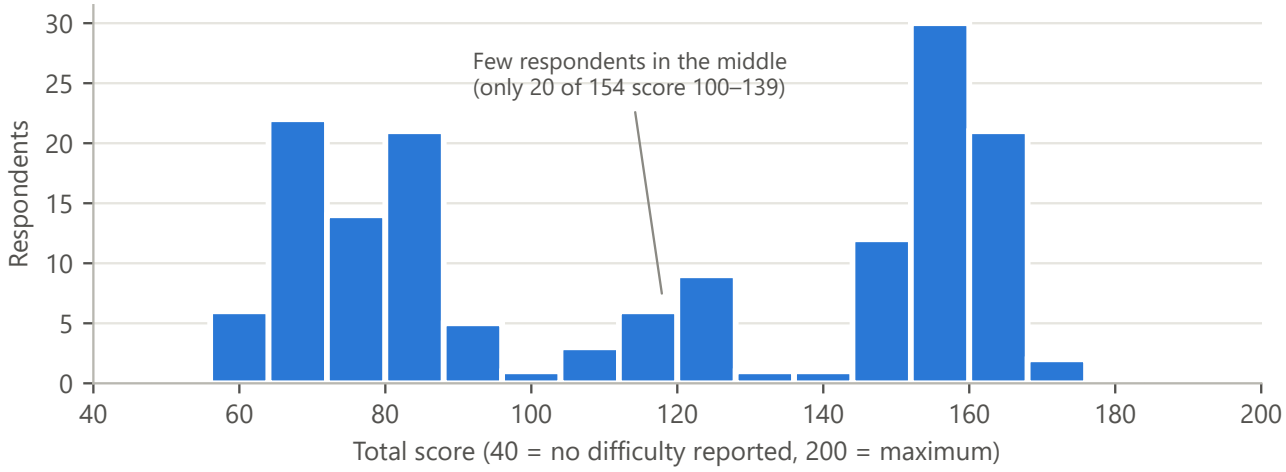
Result: Good 64, Normal 21, Mild 67, Severe 2.

The weights were discussed informally with a psychiatrist and university counsellors, but that review was not formally recorded. A sensitivity analysis tests how much the classification depends on them:

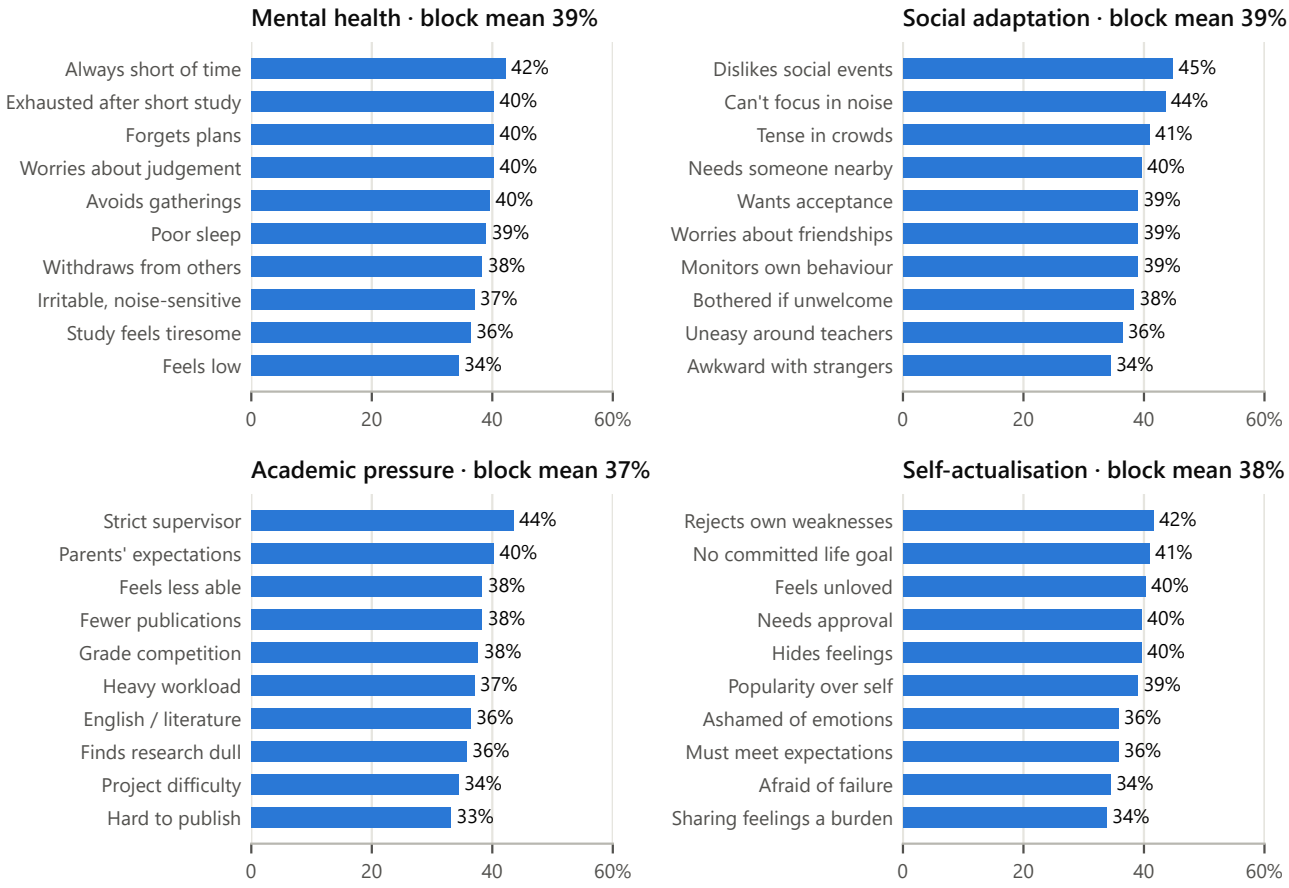


The classification hardly depends on the weights. Equal weights reproduce 94.8% of classifications (Cohen's κ = .91), and fully reversed weights still reproduce 87% (κ = .79). This follows from §4: because the subscales move together, any weighting gives almost the same result.

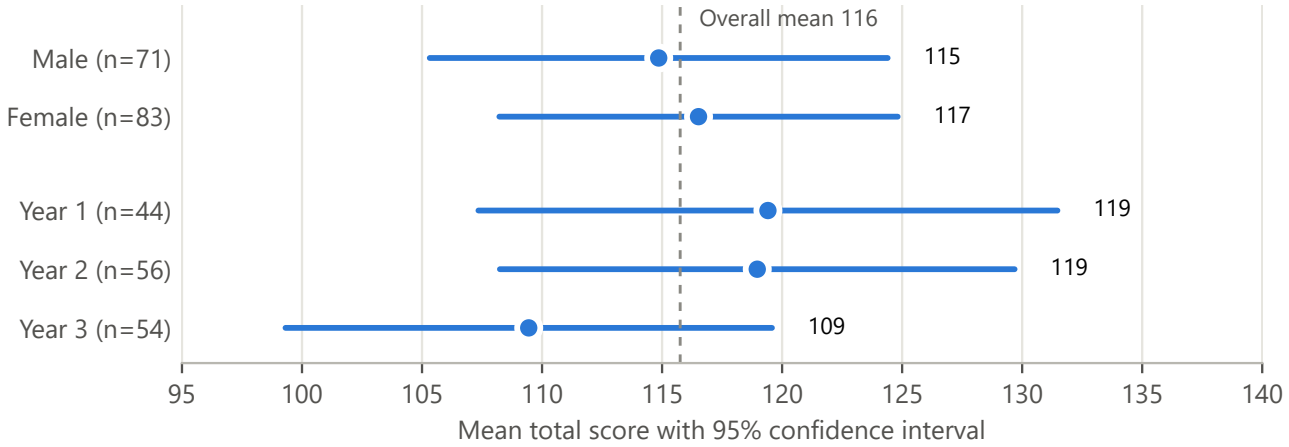
6 · Results



Total scores fall into two groups. About half of the respondents report low difficulty on almost every item and half report high difficulty, with few in between. The sample is therefore treated as descriptive, not representative (§8).



No single issue stands out. Agreement (4 or 5 on the scale) ranges only from 33% to 45% across all 40 items, and the four block averages are within 2 points of each other. The highest items are disliking social events (45%), not being able to focus in noise and a strict supervisor (both 44%). The narrow spread fits the one-factor result in §4: respondents tend to answer all items in the same direction.



No meaningful differences by gender or year of study. Gender: 115 vs 117, Welch t = 0.26, p = .79, d = 0.04. Year of study: F(2,151) = 1.09, p = .34, η^2 = .01. Third-year students score slightly lower, but the confidence intervals overlap and the effect is negligible, so it is not reported as a finding.

Claim removed from the thesis

The thesis reported that the share of students rated "Good" grew from 41.6% to 50%. That compared the full sample (n = 154) with a 10-case model test set, and a survey taken at one point in time cannot show change in any case. I have removed this claim.

7 · Can the questionnaire be shorter?

Since the 40 items mostly measure one thing, I tested whether 8 items (2 per block) give the same result. The sample was split into two random halves; the 2 items per block with the highest item–total correlation were selected in one half, and the short form was checked against the full questionnaire in the other. The procedure was repeated 200 times.

8-item short form vs. full 40 items (held-out half)	Mean	5th–95th percentile
Correlation with the 40-item total	.96	.94–.97
Cronbach's α of the 8 items	.89	.87–.91
Same level as the full scoring model	87%	81–92%
Agreement beyond chance (Cohen's κ)	.79	.69–.87
Baseline: always predict the most common level	*40%*	*34–46%*

The items chosen most often were A2, A9, B1, B3, C7, C9, D1 and D2. An 8-item version would cut the questionnaire by 80% and lose very little information *in this sample*. Before using it I would replace A9 (double-barrelled, §2) and confirm the result on a new sample, because the two-cluster distribution makes agreement easier to achieve than it would be in a typical population.

8 · Limitations and changes I would make

Limitation	Effect	What I would do in a new round
No pilot	Wording and discrimination problems only showed up after fielding	Cognitive interviews with 5–8 students, then a soft launch (n ≈ 30) and an item-statistics review before the full launch
Exclusion rules not recorded	The cleaning step cannot be reproduced	Set rules before fielding: attention check, speeders (< ⅓ of the median completion time), straight-lining, duplicates
All items scored in one direction	Acquiescence may inflate correlations between dimensions	Balanced item direction, plus one instructed-response item
Dimensions not distinct	Subscale scores cannot be interpreted separately	Write items from precise construct definitions, run EFA on pilot data, drop cross-loading items, and test discriminant validity before launch
Convenience sample with two clusters	Percentages cannot be generalised	Define the population, set quotas by year and discipline, and report results as descriptive
Expert weighting not formalised	The weights cannot be reproduced	Two-round Delphi with 5–7 experts plus AHP pairwise comparison; accept the weights at Kendall's W ≥ .5 and a consistency ratio < .10; always add a sensitivity check (§5)

Methods and tools

Stage	Methods	Tools
Design	Construct definition, item adaptation from validated scales, 5-point Likert scale	Wenjuanxing
Data quality	Missing data, straight-lining, long-string, Mahalanobis distance	Python (pandas, SciPy)
Items and reliability	Corrected item–total r, α-if-deleted, critical ratio, Cronbach's α, composite reliability	SPSS, Python
Validity	KMO and Bartlett's test, parallel analysis, EFA (oblimin), CFA, AVE, Fornell–Larcker criterion, comparison of 1-factor, 4-factor and bifactor models	AMOS, Python (factor_analyzer, semopy)
Scoring	Fuzzy comprehensive evaluation; weight sensitivity (equal, entropy, random Dirichlet weights); Cohen's κ	Python, Excel
Short form	Split-half item selection repeated 200 times, logistic regression, agreement and κ against the full scale	Python (scikit-learn)
Segment comparison	Welch t-test, ANOVA, Kruskal–Wallis, χ^2 cross-tabs, Cramér's V, effect sizes, 95% CIs	Python, SPSS

All numbers in Part A can be reproduced from the raw data file with the scripts in the GitHub repository (github.com/Newnew-niu/survey-instrument-validation).

Appendix A · The 40-item questionnaire (English translation)

The original is in Chinese. Scale: 1 = strongly disagree … 5 = strongly agree. Higher scores indicate more difficulty.

A · Mental health

A1	I always feel short of time and rush everything, e.g. crossing on a red light, walking and talking fast.
A2	I often can't remember today something I planned yesterday.
A3	I feel low for no clear reason and often sit staring blankly.
A4	I no longer enjoy gatherings with friends as much as I used to; socialising feels like a duty.
A5	I am unusually sensitive to pollution and noise, crave quiet and lose my temper easily.
A6	I don't want to go into the lab or office; studying feels tiresome.
A7	I worry about doing badly and care too much about how others judge my work.
A8	I avoid facing classmates and supervisors and tend to withdraw.
A9	After only a short time studying I feel exhausted and short of breath.
A10	I often can't sleep at night, and when I do sleep I dream constantly; my sleep is poor.

B · Social adaptation

B1	If other people don't seem to welcome me, it bothers me.
B2	Deep down, I often want other people to accept me.
B3	I try hard to watch my behaviour so that others won't avoid me.
B4	I need to feel there is always someone nearby who can help me.
B5	Around strangers I often have nothing to say and feel awkward.
B6	If I lose touch with a close friend for a long time, I worry about our relationship.
B7	In noisy, chaotic surroundings I can't concentrate and study inefficiently.
B8	I don't like social events and don't see them as a chance to make friends.
B9	The more crowded a place is, the more tense I feel.
B10	When a teacher is present, I feel ill at ease.

C · Academic pressure

C1	Many courses and a heavy workload put me under a lot of pressure.
C2	Having published less than others puts me under a lot of pressure.
C3	My English makes reading international literature hard, which puts me under a lot of pressure.
C4	I find study and research dull.
C5	Publishing papers is hard for me, which puts me under a lot of pressure.
C6	Feeling less able than others puts me under a lot of pressure.
C7	Classmates competing over grades puts me under a lot of pressure.
C8	The difficulty of my research project puts me under a lot of pressure.
C9	My supervisor's strict demands put me under a lot of pressure.
C10	My parents' high expectations put me under a lot of pressure.

D · Self-actualisation

D1	I feel ashamed of my emotions.
D2	I strongly need others to approve of what I do.
D3	Expressing negative feelings to people I love feels like a burden.
D4	I don't accept my own weaknesses.
D5	I would rather be popular than be myself.
D6	I'm afraid to express my feelings in case it leads to consequences I don't want.
D7	I feel I have to do what others expect of me.
D8	I have no life goal I am deeply committed to.
D9	I feel unloved because I am unable to give love.
D10	I am afraid of failure.

Profile

K1	Gender
K2	Year of study (1 / 2 / 3)

Appendix B · GA-BP neural-network model (thesis chapter 4)

The thesis also included a model to predict each respondent's level from the scoring results. I chose a neural network after a literature comparison of SVM, random forest, XGBoost, naive Bayes and neural networks on accuracy, complexity, robustness and scalability. To reduce the BP algorithm's tendency to get stuck in local minima, a genetic algorithm set the network's starting weights.

Setup

Component	Specification
Data	154 cases. Inputs: the four level memberships from the scoring model (Good, Normal, Mild, Severe). Output: level 0–3. Normalised, then split 80/20 at random into 124 training and 30 test cases.
Network	Three layers, 4–9–1. Hidden layers of 2 to 15 neurons were compared over repeated runs on MSE, MAE and accuracy; 9 gave the lowest error.
Genetic algorithm	Each chromosome encodes all 55 weights and biases (36 + 9 + 9 + 1), initialised in [−1, 1]. Population 30, 30 generations, MSE-based fitness, tournament selection, single-point crossover (rate 0.7), random mutation (rate 0.1).
Fine-tuning	Loaded the best chromosome into `feedforwardnet` and continued BP training, with early stopping after 6 validation failures.
Result	Test MSE = 0.0097. The GA-BP network tracked the true levels more closely than the BP network without GA initialisation.

Hidden-layer selection

Hidden neurons	2	4	6	7	9	10	12	15
MSE	.0353	.0046	.0008	.0587	**.0000**	.0065	.0019	.0337
MAE	.1248	.0320	.0158	.1873	**.0009**	.0287	.0233	.0994

Review (2026)

On review, the model predicts something already known. The output level is defined as the input membership with the highest value, so simply taking the largest input reproduces the level for all 154 cases (100%). The near-zero errors show that the network learned this rule, not that it predicts anything new. In addition, the number of hidden neurons was chosen on the same test set used to report performance, which makes the reported error optimistic.

The appendix documents the modelling and optimisation work. For a real prediction task, I would:

- use inputs measured independently of the outcome;
- tune on a validation set or with cross-validation, and keep the test set untouched;
- compare the network against a simple baseline such as logistic regression.